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Examining the neural bases of spontaneous mental experiences with real-time fMRI

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Abstract

In daily life, people are often pre-occupied by mental experiences that arise involuntarily, such as mind-wandering and rumination. In prior studies, such experiences and their neural correlates have been captured using online thought probes at random intervals during functional Magnetic Resonance Imaging (fMRI). However, this approach results in variability in temporal stages across sampled thoughts, leading to imprecise estimates of brain activity relative to the arising of mental experience. Here we aim to improve temporal precision using a novel method termed real-time fMRI-triggered experience-sampling (rt-fMRI-ES). We will analyze blood-oxygenation-level-dependent (BOLD) signals in real time during a wakeful resting state to identify spontaneous brain events that will automatically trigger thought probes. We will target increased BOLD activation in two regions of interest, the dorsal anterior insular cortex (daIC) and posteromedial cortex (PMC), based on their theorized roles in mind-wandering and experiences that occur during rumination and worry. Thought probes will additionally be triggered independent of brain activity if a significant amount of time has passed since the previous thought probe (termed “time-out trials”). H1: We predict that subjective ratings of arousal time-locked to daIC-activation trials will be higher than ratings time-locked to non-daIC-activation trials (including both PMC-activation trials and time-out trials). This hypothesis is rooted in evidence for a role of the daIC in assigning salience to spontaneous thoughts and in physiological arousal. H2: We predict that subjective ratings of attention to inner mental experience time-locked to PMC-activation trials will be higher than ratings time-locked to non-PMC-activation trials (including both daIC-activation trials and time-out trials). This is supported by a theorized role of the PMC in stimulus-independent thought, regardless of specific content. Overall, our study has potential to validate the rt-fMRI-ES technique

by demonstrating its ability to efficiently capture neural activity that is time-locked to the arising of self-generated thoughts.

1. Introduction

Much of a typical individual's mental life is characterized by involuntary thoughts, such as mind wandering and rumination, that occur independently of external stimuli (Kane et al., 2007, 2017; Killingsworth & Gilbert, 2010; Watkins & Roberts, 2020). There is growing scientific and clinical interest in understanding how the brain spontaneously generates such unprompted, inner mental experiences (Christoff et al., 2016; Kam et al., 2022; Kucyi et al., 2023; Smallwood et al., 2021). Functional neuroimaging studies have established a key role of the default mode network (DMN), and its interactions with other large-scale networks, in mind-wandering, spontaneous thought, and rumination (Andrews-Hanna et al., 2014; Berman et al., 2011; Mittner et al., 2016). Recently, involuntary thoughts have been measured during fMRI scans using online experience-sampling, where participants repeatedly provide ratings of their internal mental experiences at random time points throughout a scan. This method has yielded insights into the contents and dynamics of a wide array of ongoing, stimulus-independent experiences and their neural correlates (Christoff et al., 2009; Kucyi et al., 2013, 2021; Sormaz et al., 2018; Stawarczyk et al., 2011). However, it remains unknown how precisely ongoing mental events map on to distinct time points in spontaneous brain activity (Gonzalez-Castillo et al., 2021; Kucyi, 2018).

One of the major limitations of neuroimaging studies using the online experience-sampling method is the delivery of thought probes at random time intervals. This approach only captures a sliver of the abundance of unprompted mental experiences that an individual is bound to have during a resting state or task performance conditions. As random-interval experience-sampling is non-targeted in its execution, it has not aided in the investigation of when exactly a thought

trajectory begins and ends, and it can often miss the occurrence of specific thought subtypes such as high-arousal, distressing mental experiences that may be of interest in the context of mental health. Experience-sampling at random intervals leads to high variability in the temporal stages of sampled mental experiences, which becomes a source of noise that necessitates large sample sizes so that brain-experience relationships can be detected with group-averaging procedures that can obscure individual-level effects (Kucyi et al., 2024). An alternative “self-caught” experience-sampling method has been tested during fMRI wherein individuals to report the spontaneous arising of a thought (Ellamil et al., 2016; Hasenkamp et al., 2012). However, that approach only captures thought onsets that individuals are aware of and does not guarantee a uniform delay across trials or individuals between self-report and the brain activity that represents thought onset.

To address these critical limitations, in this study we will develop **real-time fMRI-triggered experience-sampling** (rt-fMRI-ES), a method for efficiently sampling self-generated thoughts based on spontaneous neural events. Using real-time analysis of incoming fMRI data during acquisition, we will detect instances of high blood-oxygenation-level-dependent (BOLD) activation to trigger the appearance of thought probes immediately following neural events of interest. This approach aims to match the temporal stage of arising thoughts across trials and to efficiently target thoughts that have specific qualities, such as distressing experiences that underlie maladaptive disordered thought. Moreover, by treating brain activity as an independent variable, our approach aims to empower the prospective testing of brain-experience relationships. As the current study is aiming to validate this novel rt-fMRI-ES approach, we will additionally include trials that will not be triggered by brain activity but will be displayed after ample time has passed since the previous trial (termed “time-out trials”). This is to ensure that, in the event that an

insufficient number of trials are triggered by brain activity in a given run, thought ratings are still being collected from subjects.

We will develop and validate rt-fMRI-ES by targeting both the **dorsal anterior insular cortex** (daIC), a core region of the brain's salience (or cingulo-opercular) network (Seeley et al., 2007; Uddin et al., 2023), and the **posteromedial cortex** (PMC), a component of the DMN (Buckner & DiNicola, 2019; Raichle et al., 2001). The daIC is hypothesized to assign salience to self-generated experiences (Christoff et al., 2016), and its spontaneous activation has shown to be co-occur with neurophysiological markers of arousal (Kucyi & Parvizi, 2020; Schneider et al., 2016). Coupled with its atypical structure, function, and connectivity in psychiatric illness at a transdiagnostic level (Goodkind et al., 2015; Sha et al., 2019), there is a need to understand the role the daIC plays in characterizing distressing self-generated thought. In line with evidence of the daIC's role in salient experiences, we predict that subjective ratings of arousal time-locked to daIC-activation trials will be higher than ratings time-locked to non-daIC-activation trials (including both PMC-activation trials and time-out trials) (*H1*). The PMC, which includes subregions of posterior cingulate cortex and precuneus, activates during internal mentation and self-related mental representations, and deactivates during explicit goal-directed attention states (Buckner & DiNicola, 2019; Cavanna & Trimble, 2006). Given the PMC's role in self-referential and stimulus-independent thought, regardless of specific content, we predict that subjective ratings of attention to inner mental experience time-locked to PMC-activation trials will be higher than ratings time-locked to non-PMC-activation trials (including both daIC-activation trials and time-out trials) (*H2*). By targeting two regions of interest (ROIs) that often show anticorrelated activity with one another (Dixon et al., 2017; Fox et al., 2005), our study has potential to validate

the rt-fMRI-ES technique by demonstrating its ability to efficiently capture neural activity that is time-locked to the arising of self-generated thoughts involving distinct phenomenology.

2. Methods

Overview

The current study will be comprised of two sessions taking place at the Temple University Brain Research and Imaging Center: (1) An initial 1-hour visit consisting of consent documentation, a study agenda overview, task training, and self-report questionnaires; and (2) The main 2-hour visit consisting of brief self-report questionnaires, task training recap, and 90 minutes of MRI procedures (see **Fig 1a**). During the MRI procedures, we will collect localizer scans, fieldmap scans, and a T1-weighted anatomical scan prior to fMRI scanning. While undergoing fMRI scanning, participants will complete an 8-minute run of the Gradual-onset Continuous Performance Task (GradCPT), which is our functional localizer task (see section *GradCPT* for more information) that will be used to define ROIs within individuals. Participants will then undergo an 8-minute eyes-open resting state fMRI scan while we will compute and visually inspect individual-level ROIs from functional localizer data. This resting state scan will not be used in the final analysis. We will then use the individual-level ROIs for our main experimental task, the real-time fMRI-triggered experience sampling (rt-fMRI-ES) paradigm (6 runs, 8-minutes each), in which experience sampling trials (see **Fig 1b**) will be triggered based on BOLD activation level in a preselected ROI. The ROI for each rt-fMRI-ES run will be selected with a counterbalanced order, yielding 3 runs targeting BOLD events in the daIC and 3 runs targeting BOLD events in the PMC. The feasibility of our pipeline was established using a pilot subject who successfully completed all components of both sessions of the study.

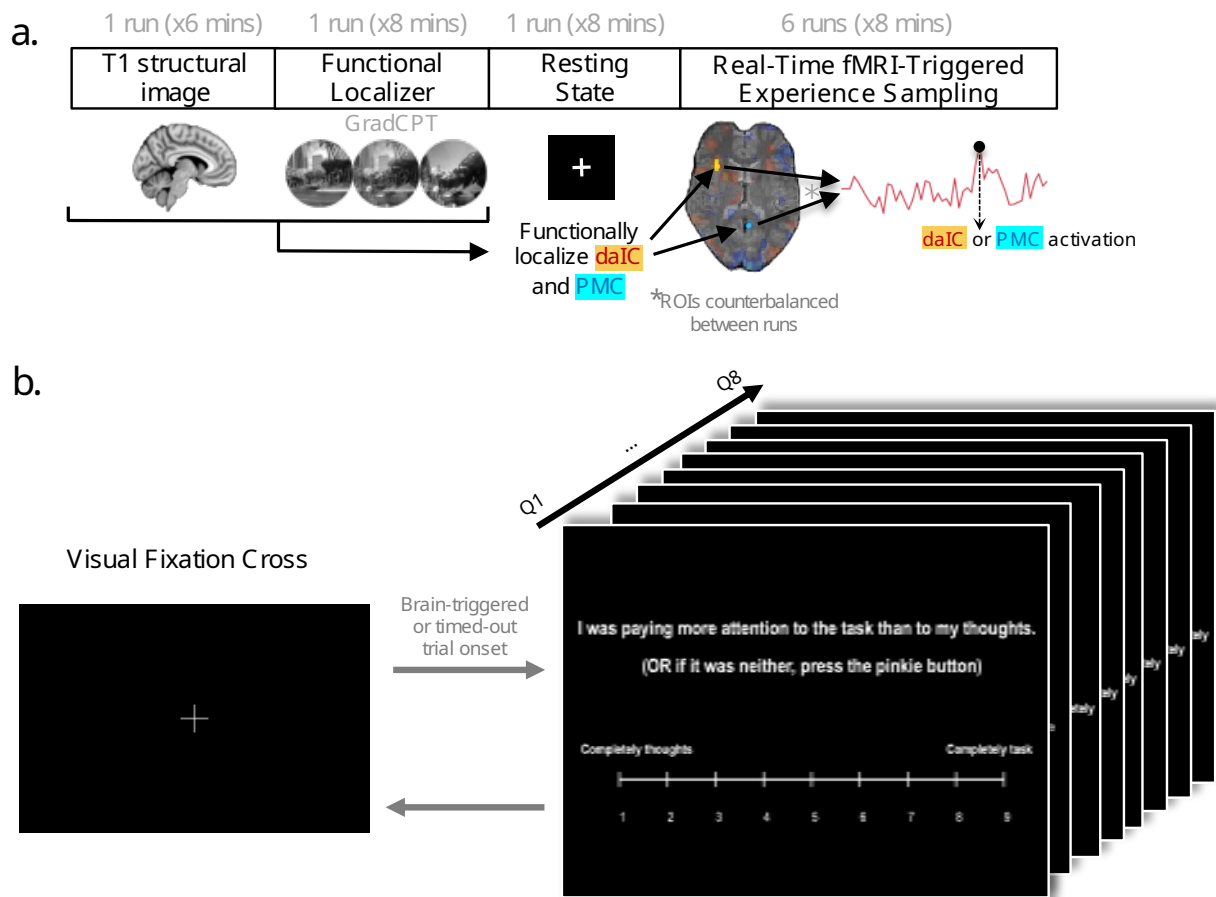


Figure 1. Overview of study procedures and task. a) Within a single neuroimaging session, participants will undergo a T1 structural scan and a functional MRI scan while performing the Gradual-onset Continuous Performance Task (GradCPT). Stimulus timing from the GradCPT will be used in conjunction with the participant's structural and task fMRI images to compute individually localized regions of interest (ROIs) during a resting-state fMRI run. Subsequently, across six runs of a real-time fMRI condition, brain activity within these individualized functional ROIs, the dorsal anterior insular cortex (daIC) and the posteromedial cortex (PMC), will be analyzed in real-time, with ROIs being counter-balanced across runs. Experience sampling trials will be triggered based on increases in BOLD activation within the targeted ROI. b) Within each run of real-time fMRI-triggered experience sampling, participants will view a visual fixation cross at the center of the screen. Presentation of the visual fixation cross will be interrupted by trials of thought probes triggered by increases in BOLD activation in the run ROI, or by lack of brain-triggered trials after 100 seconds ("time-out" trials). Each experience sampling trial will consist of 8 thought probe items with 9-point Likert scales. Following the completion of an experience sampling trial, the visual fixation cross will return on the screen.

Participants

Sixty adults will be recruited from the larger communities of Drexel University and Temple University in Philadelphia, PA, through study flyers and word of mouth (age range: 18 – 35, 30 male, 30 female, 28 racial and/or ethnic minority, 32 Caucasian/White). Participants will be right-handed, fluent in English, free of any internal or external metallic objects contraindicated by a 3 Tesla magnetic field, have normal or corrected-to-normal vision, and have no history of a psychiatric, neurological, or chronic medical condition. Eligibility will be assessed through a REDCap (Harris et al., 2009) pre-screening survey with questions about study-specific inclusion criteria and a subsequent telephone screener in which a study team member will verbally review an MRI Subject Safety Screening Form. Participants will be paid for their participation at a rate of \$35 USD per hour. All study protocols have been approved by the Drexel University Institutional Review Board (IRB).

Procedures

Day 1 Session

Following consent acquisition, participants will begin with study task training. During training for GradCPT, the task will be colloquially described to the participant as the “Cities and Mountains Task.” Following instruction review, participants will be presented with a 1-minute demo version of GradCPT to complete as practice. During task training for rt-fMRI-ES, the main objective of the task will be framed as an attention task in which participants are to try to direct their attention to a fixation cross during the entirety of the run. Participants will be informed that once every minute or so, the fixation cross will be replaced by a series of 8 questions that will ask about the content, quality, and dynamics of any thoughts that they were having directly preceding

the appearance of the question set. To avoid performance bias, participants will not be made aware of the time congruency of the thought probes relative to brain activity; timing of thought probe appearance will be framed as being random. The definitions and scale anchors for each thought probe will be thoroughly discussed. Following this, participants will be instructed to complete a single-trial demo of the rt-fMRI-ES paradigm in which they are to rate responses to each question based on an example thought that had been read aloud by the researcher. This exercise is to gauge participant understanding of the task at hand, as well as the meaning of each thought probe. Upon completing the demo, the researcher will have participants verbally describe how they responded for each question and will correct any misunderstandings.

Participants will privately complete a battery of mental health and behavioral questionnaires on REDCap in a testing room. This survey will consist of a section inquiring the participant's legal full name and municipality of birth as it appears on their birth certificate to adhere to the National Institute of Mental Health National Data Archive Global Unique Identifier requirements (<https://nda.nih.gov/>). Questionnaires in this survey will include the DSM-5-TR Cross-Cutting Symptom Measure – Adult (American Psychiatric Association, 2013), GAD-7 Anxiety (Spitzer et al., 2006), Patient Health Questionnaire 9 (Kroenke et al., 2001), Ruminative Response Scale (Nolen-Hoeksema & Morrow, 1991), STAI – Trait (Spielberger et al., 1983), World Health Organization Disability Assessment Schedule (Ustun et al., 2010), Perceived Stress Scale (Cohen et al., 1983), Adult ADHD Self-Report Scale (Kessler et al., 2005), Perseverative Thinking Questionnaire (Ehring et al., 2011), and the Mind Wandering Deliberate-Spontaneous Scale (Mrazek et al., 2015). Data from self-report questionnaires will not be used in the pre-registered analyses reported here.

Day 2 Session

On the second visit of the study, participants will begin with a 45-minute preparation session where they will complete an MRI Subject Safety Screening Form, a behavioral-state questionnaire, a review of the scanning agenda, and completion of a second review of task training for GradCPT and rt-fMRI-ES paradigms. The questionnaires will consist of the STAI – State (Spielberger et al., 1983) and the Stanford Sleepiness Scale (Hoddes et al., 1972) to assess the participant's current state of anxiety and alertness. Metal checks will be performed with a metal detecting wand, and a ferromagnetic metal detecting pillar. The participant will be set up on the MRI bed with the button box placed in their dominant hand to make responses during the tasks. An emergency squeeze ball will be placed in the participant's other hand for them to squeeze in case they need immediate attention during the session. When squeezed, an alarm would sound in the control room, alerting research personnel to attend to the participant.

In the scanner control room, we will first map the shared folder intended for incoming DICOMs (raw scanner images) on our project laptop to the Siemens console Y drive (see *Real-time fMRI Analysis* section). Once the folder is mounted, we will use the *idealcmdtool* package (a tool that is installed on all Siemens consoles) to enable sharing of incoming images with whichever external drive is currently mapped. Within the shared folder, we will create a subject folder for running our functional localizer package, rtfMRI-prep (see *Functional Localizer Analysis* section). We will begin the session with a localizer scan, Auto-Align scout scans, and an aligned localizer scan. Following the T1 image acquisition, a B0 field map scan will be run. Immediately following completion of the GradCPT fMRI task scan, our GradCPT MATLAB code will output a text file that labels the timing of each mountain event in the participant's run. We will transfer this text file to the subject folder created in the beginning of the session. In a Terminal window, we will use

220 *dcm2niix* (Li et al., 2016) to convert all existing DICOMs in the shared folder into NIFTI files, as
221 required by our functional localizer package. NIFTI files of the participant's T1-weighted
222 structural image and GradCPT BOLD fMRI data will be transferred into the subject folder. While
223 the resting state fMRI scan is being conducted, we will run our functional localizer package (see
224 *Functional Localizer Analysis* section) from the subject folder containing the GradCPT event text
225 file, the T1-weighted structural image, and the GradCPT BOLD fMRI image.

226 Once our functional localizer package is finished running, we will launch the Open
227 Neurofeedback Training (OpenNFT, v1.0.0rc0) software to begin the first run of rt-fMRI-ES.
228 OpenNFT is an open-source Python/MATLAB package that has the ability to detect incoming
229 fMRI volumes, perform basic preprocessing, processing, signal estimation and stimulus
230 presentation for a variety of real-time fMRI protocols (Koush et al., 2017). In the OpenNFT
231 graphic user interface, we will load in all necessary parameters that the software needs to perform
232 processing and stimulus presentation during the run (see *Real-time fMRI Analysis* section). For
233 efficiency, OpenNFT allows the user to input a configuration file that lists the desired settings of
234 each necessary parameter. Our functional localizer package will have created these configuration
235 files within the subject folder for each upcoming rt-fMRI-ES run. We will then begin the OpenNFT
236 session, and start the rt-fMRI-ES scan. This process will be repeated for each of the six rt-fMRI-
237 ES task runs. To ensure task engagement, we will actively check for consistent and timely
238 responses to each thought probe by observing the participant's screen from a monitor in the control
239 room during each run. We will note any signs of lack of engagement (i.e. lack of response to
240 thought probes). In between each rt-fMRI-ES run, we will verbally check in with participants over
241 an intercom to ensure that they are alert.

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Neuroimaging Data Acquisition

Neuroimaging data will be acquired using a Siemens 3-Tesla MAGNETOM Prisma MRI Scanner using a 64-channel head coil at the Temple University Brain Research and Imaging Center. Following the collection of localizer scans, a T1-weighted anatomical image will be acquired with an MPRAGE sequence (repetition time: 2400 ms, echo time: 2.22 ms, inversion time: 1000 ms, flip angle: 8°, field of view: 256 x 256mm, 208 slices, 0.8mm isotropic voxels). We will also acquire a B0 field map for correction of echo-planar imaging (EPI) fMRI images (repetition time: 789 ms, echo time 1: 4.92 ms, echo time 2: 7.38 ms, flip angle: 45°, field of view: 192mm x 192mm, 2mm slice thickness, 2mm isotropic voxels). Functional magnetic resonance imaging (fMRI) runs will be performed with a gradient EPI multiband sequence (repetition time: 2000 ms, echo time: 25 ms, flip angle: 70°, field of view: 192mm x 192mm, 2mm slice thickness, 2mm isotropic voxels, multi-band acceleration factor: 3, phase encoding direction: anterior-to-posterior) with collection of 81 transverse slices aligned with the AC-PC plane. In total, 8 fMRI runs will be collected per session for each participant (one GradCPT run, one resting state run, and six rt-fMRI-ES runs).

Experimental tasks

GradCPT

The Gradual-Onset Continuous Performance Task (GradCPT) is a continuous performance task that engages sustained attention (Esterman et al., 2013). During a run of GradCPT, participants view a series of city- and mountain-scene grayscale images that quickly and continuously transition from one image to another every 800 ms (using linear pixel-by-pixel interpolation between consecutive images). Scene images are presented randomly, with city scenes

frequently visualized (~90% of total trials) and mountain scenes being displayed infrequently (~10% of total trials), while the same scene cannot repeat on consecutive trials. Participants are instructed to press the index finger button on an MR-safe Current Designs 4-finger button box for every frequent trial (city scenes), and to withhold button presses for infrequent trials (mountain scenes). Performance can be assessed through analysis of reaction time, as well as accuracy.

Prior fMRI and intracranial electrophysiology studies have shown highly reliable and replicated salience network (including daIC) activation and DMN (including PMC) deactivation in response to infrequent stimuli (i.e., mountains) in the GradCPT (Esterman et al., 2013; Fortenbaugh et al., 2018; Kucyi et al., 2020; Kucyi & Parvizi, 2020). As such, GradCPT will serve as a functional localizer task to obtain the current study's regions of interest (daIC and PMC) within each individual participants' native brain space. Participants will complete a single 8-minute run of GradCPT, which will be displayed using Psychophysics Toolbox in MATLAB (Brainard, 1997). Upon completion of GradCPT, the participant's commission error rate (rate of erroneous button presses to infrequent mountain images) and omission error rate (rate of missed button presses during frequent city images) will be calculated and displayed to the experimenter in the MATLAB Command Window. The GradCPT stimulus timing and fMRI data will be used as input for an in-house functional localizer package, *rtfMRI-prep*, that generates individual-level regions of interest in daIC and PMC using a general linear model analysis (see *Functional Localizer Analysis* section). If the participant demonstrated inadequate performance effort during the task, as shown by a commission error rate of 0.75 or an omission error rate of 0.33, an alternative script within *rtfMRI-prep* that generates standard space regions of interest rather than individual-level regions will be used. While these GradCPT performance error rates are rather high (more than 5 times worse than the average performance in healthy adults; Esterman et al., 2013;

Fortenbaugh et al., 2018), we intentionally set a liberal threshold as we are only trying to capture subjects who are not performing at all, rather than individuals who have poor performance ability. While we have an alternative, standard-space ROI option in place in the event of GradCPT performance being insufficient for use in functional localization, we do not anticipate that participants will not engage in, or fall asleep during, the GradCPT task.

Resting State

Following performance of the GradCPT, participants will complete an 8-minute resting state fMRI scan. They will view a central fixation cross and will be instructed to keep their eyes open to prevent drowsiness. Participants will not make any task responses during this run. The fMRI data from this resting state run will not be used for preregistered analysis. This scan purely serves as an idle grace period to compute and visually inspect individual-level daIC and PMC ROIs (see section *Functional Localizer Analysis*), and to initialize the real-time fMRI analysis software that we will use for our rt-fMRI-ES task runs (see section *Real-time fMRI Analysis*).

Real-time fMRI-triggered experience sampling (rt-fMRI-ES)

We developed a novel experience-sampling paradigm, the real-time fMRI-triggered experience sampling (rt-fMRI-ES) task that we will use to sample participants' spontaneous mental experiences during six 8-minute fMRI runs. While this paradigm will be used in conjunction with our real-time fMRI system to trigger thought probes based on brain activity, it will be framed to participants as a visual fixation task with thought probes occurring randomly to prevent potential performance bias. In addition to triggering the appearance of the paradigm based on brain activity in either ROI, the paradigm will appear on the screen if no significant brain activation events have been detected by the real-time system after a predefined time window

(termed “time-out” events / trials). Participants will be instructed to focus their attention on a fixation crosshair and will be informed that a set of 8 rating scales (a single trial) will appear at random times for them to report the qualities of thoughts that occurred immediately preceding the appearance of the trial (see *Real-time fMRI Analysis* for description of trial timing). Each rating scale will be accompanied by a 9-point Likert scale with an arrow visualized on the center of the scale. Participants will use an MR-safe Current Designs 4-finger button box to make responses. The middle finger button will move the arrow left, the ring finger button will move the arrow right, and the index finger will submit participant responses to each scale. Only for the first rating scale of each set, there will be an option to use the pinkie finger button to report that the participant was neither paying attention to the central fixation cross nor their internal mental thoughts. This option is included to distinguish thoughts that fall on some continuum of attention to task performance versus ongoing mental experience, from moments of external distraction or mind-blanking. Distraction/mind-blanking trials will be excluded from analyses (see section *Criteria for Excluding Data*). The rating scales, along with their anchors, are listed below:

Q1. Attention: “I was paying more attention to the task than to my thoughts. (OR if it was neither, press the pinkie button)”

- Anchors: **1 = completely thoughts, 9 = completely task, or pinkie button = neither thoughts nor task**

Q2. Affect: I felt very positive.

- Anchors: **1 = extremely negative, 9 = extremely positive**

Q3. Arousal: I felt very activated / energized.

- Anchors: **1 = Not at all, 9= Completely**

Q4. Perceived repetitiveness: This thought was similar to those I have had before.

- Anchors: **1 = Not at all, 9 = Completely**

Q5. Movement: My thoughts were freely moving.

- Anchors: **1 = Not at all, 9 = Completely**

Q6. Engagement: My thoughts were difficult to disengage from.

- Anchors: **1 = Not at all, 9 = Completely**

Q7. Rumination: I was ruminating about the past.

- Anchors: **1 = Not at all, 9 = Completely**

Q8. Worry: I was worrying about the future.

- Anchors: **1 = Not at all, 9 = Completely**

Rating scales will largely be self-paced, with a 12-second time-out if response submission to each rating scale has not yet been made. Following the completion of each trial (a full set of the 8 rating scales above), the visual fixation cross will reappear on the screen. Only thought probe items Q1 and Q3 will be used for preregistered analyses (see section *Statistical Analyses*). The remaining thought probe items will be collected for purposes that are beyond the scope of this preregistration. Any exploratory analyses regarding these remaining thought probes will be included in the final manuscript and will be labeled as exploratory. None of the conclusions drawn from these exploratory analyses will affect the overall interpretation of the main findings specified by our pre-registered hypotheses.

3. Data Analyses

Functional Localizer Analysis (rtfMRI-prep)

We developed an in-house functional localizer package, *rtfMRI-prep* (https://github.com/DynamicBrainMind/rtfMRI_prep), which will be used to identify the daIC and the PMC within the participants' brains. The package will be run promptly after collection of the T1-weighted anatomical scan and GradCPT fMRI scan, which will be immediately accessible from our project laptop through real-time data sharing with the Siemens scanner console (see *Day 2 Session*). The script within *rtfMRI-prep* that we intend to use during every neuroimaging session is *rtfmri_localizer_proc*, which identifies the daIC and the PMC at the individual level based on participants' BOLD activation during task-related events in the GradCPT. However, if a participant demonstrated inadequate performance effort during the GradCPT, as indicated by a commission error rate of 0.75 or an omission error rate of 0.33 (see section *GradCPT*), we will use an alternate script within the package, *rtfmri_localizer_proc_noGLM*, to generate standard space regions of interest. We do not anticipate needing to use *rtfmri_localizer_proc_noGLM*, as we expect that it will be very rare that participants will not engage in the GradCPT task. Using the participant's anatomical scan, GradCPT fMRI scan, and a 3-column text file detailing GradCPT task event information, *rtfMRI-prep* will perform basic fMRI preprocessing and compute functionally localized ROIs using a combination of FSL v6.0.7.7 (Jenkinson et al., 2012). functions in approximately 4.5 minutes on a high-powered Apple 14-inch Macbook Pro (Apple M3 Max Chip with 14 cores). The processes executed by either script within the *rtfMRI-prep* package are detailed below.

374 Individual-level regions of interest (rtfmri_localizer_proc)

375 Using *rtfmri_localizer_proc*, motion correction will first be performed on fMRI data using
376 *MCFLIRT* (Jenkinson et al., 2002) with 6 transform degrees of freedom to align all volumes to the
377 middle volume, followed by brain extraction of fMRI data using *BET2* (Jenkinson et al., 2005)
378 with a fractional intensity threshold of 0.3. The anatomical image will be skull stripped with *BET*
379 (using robust brain center estimation), and we will use *FLIRT* (Jenkinson & Smith, 2001) to
380 perform linear registration (6 degrees of freedom, trilinear interpolation) between fMRI and
381 anatomical images. We will perform linear registration between T1 and MNI152 standard space
382 using *FLIRT* (12 degrees of freedom, trilinear interpolation). Then, the obtained transformation
383 matrices will be concatenated to transform from the functional space to standard space. We will
384 immediately visually inspect the quality of image registrations using FSL's *slices* tool. We will
385 then perform spatial smoothing of the brain-extracted functional image using *FSLMATHS* with a
386 6-mm full width at half maximum Gaussian kernel with mean filtering. As a final preprocessing
387 step, we will perform highpass filtering at 0.01 Hz using *FSLMATHS*.

388 Using the preprocessed fMRI data, we will perform a general linear model (GLM) analysis
389 using *FEAT* (Woolrich et al., 2001) with FMRIB's improved linear model prewhitening. The GLM
390 will include one regressor for the onset of mountain (i.e., infrequent target) stimuli, convolved
391 with a gamma hemodynamic response function (mean lag = 6 seconds; standard deviation = 3
392 seconds), plus the temporal derivative. We will then search for the top activated and deactivated
393 voxels, respectively within the right daIC and bilateral PMC, based on voxelwise contrast of
394 parameter estimate values. We will constrain our search to voxels within standard (MNI152) space
395 daIC and PMC masks obtained from publicly shared fMRI-derived functional parcellations that
396 delineated these regions (see **Fig 2a**). For the daIC, we will use cluster number 2 from a previously

reported, data-driven 3-cluster insular cortex parcellation solution (Kelly et al., 2012). For the PMC, we will obtain regions labelled as DMN within the Schaefer atlas (Schaefer et al., 2018) and include only DMN regions within the PMC cluster (which includes subregions of posterior cingulate cortex, precuneus, and retrosplenial cortex). Although we use the Schaefer atlas for the PMC due its clear delineation of this region within the DMN, we will not use the Schaefer atlas for defining the daIC because the relevant clusters in this atlas are contiguous with adjacent regions in the frontal operculum that are not the focus of our hypotheses.

We will register the daIC and PMC standard (MNI152) space masks to fMRI using the previously computed linear transforms. Subsequently, we will retain the top 200 activated voxels (i.e., highest contrast of parameter estimate values) within the daIC mask and the top 200 deactivated voxels within the PMC mask to yield the final individual-level ROIs (see **Fig 2b**). Importantly, the middle fMRI volume from the GradCPT run, which will be used for motion correction during functional localizer processing, will also be used as a registration template for real-time fMRI analysis (as described in “Real-time fMRI Analysis”). As such, the individual-level ROIs will be aligned with incoming fMRI data. After creation of individual-level ROIs, we will immediately inspect functional anatomy of the daIC and PMC using FSL’s *fsleyes* tool.

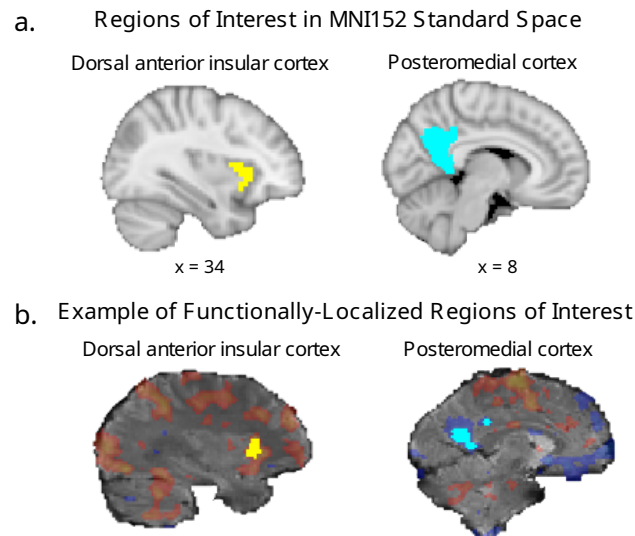


Figure 2. Regions of interest. a) Our functional localizer package, *rtfMRI-prep*, searches for the top activated and deactivated voxels, respectively, within the right dorsal anterior insular cortex (daIC) and bilateral posteromedial cortex (PMC), in response to mountain scene events (rare target images) during GradCPT performance. The search is constrained to voxels within standard (MNI152) space daIC and PMC masks obtained from publicly shared fMRI-derived functional parcellations that delineated these regions. For the daIC, cluster number 2 from a previously reported, data-driven 3-cluster insular cortex parcellation solution (Kelly et al., 2012) is shown in yellow overlaid on top of a standard space brain. For the PMC, DMN regions within the PMC cluster of the Schaefer atlas (Schaefer et al., 2018) are shown in light blue. b) The *rtfMRI-prep* package generates whole-brain voxel-wise activation and deactivation maps depicting brain activity during the subject's GradCPT run, in addition to the final individually-localized ROIs that will be used for rt-fMRI-ES runs. The individually-localized daIC (yellow) and PMC (light blue) of a pilot subject who completed the full protocol of the current study are shown overlaid on top of a whole-brain activation (red) and deactivation (dark blue) map visualized in *fsleyes*.

Standard space regions of interest (*rtfmri_localizer_proc_noGLM*)

Using *rtfmri_localizer_proc_noGLM*, the anatomical image will be skull stripped with *BET* (using robust brain center estimation), and we will use *FLIRT* (Jenkinson & Smith, 2001) to perform linear registration (6 degrees of freedom, trilinear interpolation) between fMRI and anatomical images. We will perform linear registration between T1 and MNI152 standard space using *FLIRT* (12 degrees of freedom, trilinear interpolation). Then, the obtained transformation

matrices will be concatenated to transform from the functional space to standard space. We will immediately visually inspect the quality of image registrations using FSL's *slices* tool.

Using FLIRT, we will transform standard (MNI152) space daIC and PMC sphere ROIs to the middle volume of the participant's GradCPT fMRI image. Both spheres will be obtained from publicly shared fMRI-derived functional parcellations that delineated these regions (as described in "Individual regions of interest (*rtfmri_localizer_proc*)"). We created both sphere ROIs by obtaining the center-of-gravity coordinates of both regions, then delineating 6-mm radius spheres around these coordinates. These spherical ROIs, which are similar in volume to the resulting ROIs created by *rtfmri_localizer_proc*, will then be transformed to the participant's functional space.

Real-time fMRI Analysis (OpenNFT)

Real-time analysis of incoming fMRI volumes during runs of our rt-fMRI-ES task will be conducted using OpenNFT v1.0.0rc0 software (Koush et al., 2017). OpenNFT has been successfully implemented in prior real-time fMRI studies to trigger neurofeedback and visual stimuli during behavioral performance (Pamplona et al., 2020). We will use OpenNFT on a high-powered project-dedicated laptop (Apple 14-inch MacBook Pro with an Apple M3 Max chip with 14 cores) running Python 3.6.5. and MATLAB R2020b.

We will facilitate data transfer between the MRI scanner and our project laptop via mapping a shared folder created on our laptop to the Siemens MRI console's Y drive via TCP/IP connection. Once this connection is established, the scanner will deposit incoming DICOM files (single-frame data) into the mounted shared folder. Prior to starting each OpenNFT session (each run of rt-fMRI-ES), the file path of this shared folder will be pre-specified in the OpenNFT session

parameters as the folder that the software will monitor for incoming images (termed the “Watch Folder”). Other parameters that require pre-specification include the expected first file name, the repetition time, matrix size, and voxel size of brain volumes, and file paths of the folder containing the run’s ROI, the folder that OpenNFT will save data into following completion of the run, and the folders containing template functional and structural images of the subject’s brain.

Upon acquiring each individual brain volume, OpenNFT will perform its default preprocessing steps using functions from SPM12 software (<https://www.fil.ion.ucl.ac.uk/spm/>), as described previously (Koush et al., 2017). These steps include spatial realignment to a template volume (from the GradCPT run; see above), estimation of six movement parameters (translation and rotation), reslicing, and spatial smoothing (5-mm full width at half maximum Gaussian kernel). Following OpenNFT default settings for further processing, we will perform first-order auto-regressive correction to account for temporal autocorrelation that may be due to physiological noise (Lindquist, 2008). In order to account for global signal drift, we will use several preprocessing tools. Specifically, we will use an incremental GLM (iGLM) to regress out six head motion parameters and linear trends, and a high-pass filter will be applied with a 1/128 sec cut-off. To further deal with high-frequency noise, a modified Kalman filter will be applied (Koush et al., 2012).

For the current study, OpenNFT will use an iGLM to calculate the BOLD percent-signal-change (%SC) of each incoming DICOM relative to the median BOLD baseline period, which will be set to the first 10 brain volumes (20 seconds) of the run. After the baseline period, OpenNFT will begin to screen incoming data for candidate events. Though the run ROI (daIC or PMC) being screened for events is counterbalanced across all six runs of rt-fMRI-ES, OpenNFT will screen activity in both ROIs for every run. In other words, while activity in both ROIs will be monitored

throughout the duration of each of the six runs, only one of the ROIs will be the “target region” that rt-fMRI-ES stimuli (thought probe set) will be triggered from for a given run. The second, nontarget ROI will serve as a control region. To trigger the appearance of rt-fMRI-ES stimuli based on BOLD activity in the target region of the given run, we will use an algorithm that triggers stimulus presentation whenever the following criteria are met: (1) There was a consecutive BOLD %SC signal increase across three time points (volumes) in the target region; (2) At the current volume, the BOLD %SC of the target region is greater than the run’s baseline level (i.e., greater than zero value); and (3) Criteria #1 and #2 were not met in the non-target region; (4) At least 50 seconds (i.e., 25 TRs) have elapsed since the last thought probe set was triggered.

For the first criterion, it is important to note that not all three consecutive time points need to be high; rather, we will be screening for evidence that activation in the target region is ramping up towards a peak. Our decision to trigger trials based on a successive increase across three time points in particular was motivated by simulations run on a pilot subject’s 8-minute resting data to compare the number of trials triggered when the algorithm searched for successive increases across a varying number of timepoints. The pilot subject’s resting state data was simulated using four versions of the triggering algorithm: a version that triggered a trial when there was an increase in brain activity across 2, 3, 4 and 5 TRs. Each of the four algorithms were simulated twice, one for each ROI (a run searching for activity in the daIC, and another run searching for activity in the PMC). While the versions that triggered trials based on increases across 4 TRs and 5 TRs were overly conservative (4 TRs triggered 4 PMC-triggered trials and 5 daIC trials; 5 TRs triggered 3 PMC trials and no daIC trials), the version that triggered trials based on increases across 2 TRs proved to be excessively liberal by triggering the maximum number of trials possible in both brain regions (7 trials in both runs) (Fig 3). Thus, the algorithm that triggered trials based on a successive

increase in brain activity across 3 TRs was the most reasonable option (7 PMC-triggered trials, 6 daIC-triggered trials). Moreover, upon inspecting the events that were triggered by the different algorithms, we found that the 3-TR algorithm, relative to the other algorithms, more reliably detected events that preceded or were at BOLD peaks.

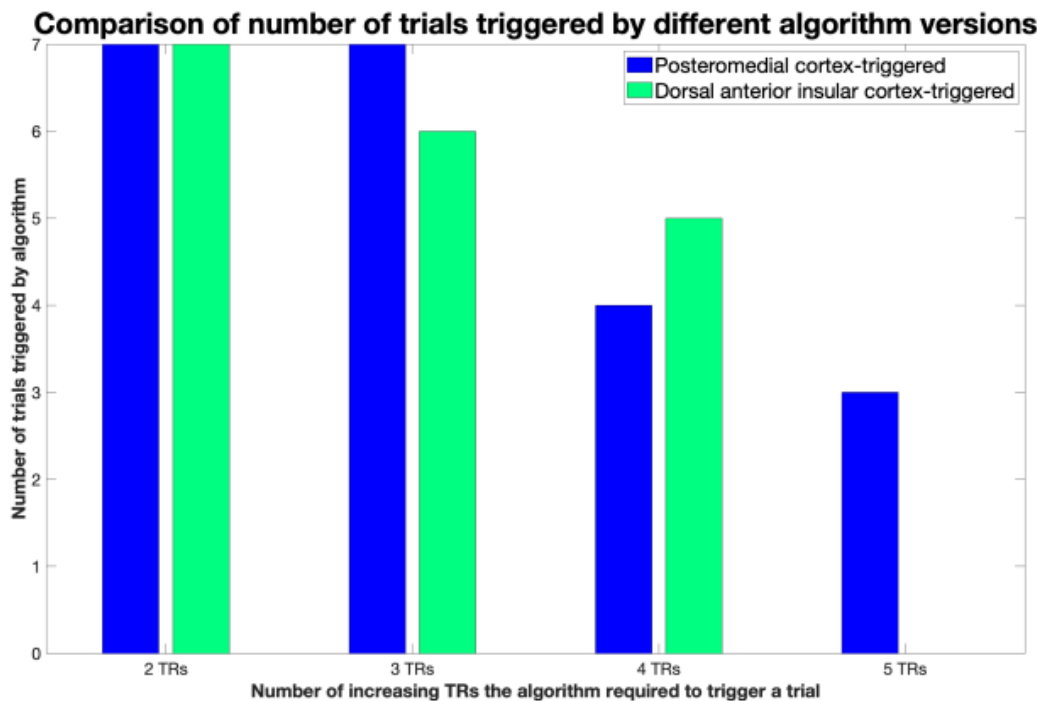


Figure 3. Comparison of the number of trials triggered by different versions of the rt-fMRI-ES triggering algorithm. To justify our decision for our triggering algorithm to present trials based on increases across three time points (3 TRs), we ran simulations on a pilot subject's 8-minute resting state data using four separate version of the triggering algorithm that triggered trials based on successive increases across a variety of time points: 2, 3, 4, and 5 TRs. Each algorithm version was simulated once each for two ROIs: one simulation searched for activity in the posteromedial cortex (PMC; dark blue bars), and another searched for activity in the dorsal anterior insular cortex (daIC; green bars).

Furthermore, even when accounting for both the 6-second delay that is consequential of the first criteria and the ~4-6 second hemodynamic delay characteristic of BOLD activity measurements, the timing of thought probe onset in the current study will be more efficient in capturing significant activity associated with mind wandering compared to traditional, randomly

519 timed experience sampling methods. Our past research shows that significant BOLD activity in
520 DMN subregions is often present and sustained as early as ~20 seconds prior to thought probe
521 onset (Kucyi et al., 2024). Therefore, though a thought probe in the current study will likely be
522 displayed about 12 seconds following the arising of a neural event of interest, this will likely be
523 far timelier in capturing this activity than typical random-onset experience sampling methods.

524 We selected our algorithm criteria based on simulations of previously collected fMRI data
525 using OpenNFT's offline session feature. By monitoring activity in both ROIs, we can ensure that
526 thought probe sets triggered by brain activity in the target region would not otherwise be triggered
527 by the non-target region. This is vital to guarantee that the neural events triggered by the two ROIs
528 are distinct. It is important to note that our decision to concurrently screen activity in both ROI
529 was made after receiving peer-review feedback on our preregistration report, which was after the
530 first 15 subjects were collected. During the rt-fMRI-ES runs of the first 15 subjects, the OpenNFT
531 software was configured to only monitor activity in the target region of each run. After changing
532 our algorithm to screen both regions, we retroactively simulated the first 15 subjects' runs to see
533 if any of their completed trials would have also been triggered by the non-target region. Across all
534 subjects' total completed trials, an average of 3 trials per subject would have also been triggered
535 by non-target regions. We will exclude these trials from the final analysis.

536 The algorithm criteria will ensure that thought probe sets do not appear with excessive
537 frequency and that triggers will be initiated when the slope of BOLD %SC in the target region is
538 increasing, thereby signifying the onset of an 'activation' event. If no significant increase is
539 detected after a considerable period of time since onset of the last thought probe set (100 seconds),
540 then the experience sampling paradigm will be automatically triggered (termed "time-out" events
541 / trials). We do not expect time-out trials to appear frequently, though in the event that a subject

has a considerable amount of such trials, they will be accounted for in our statistical models (see section *Statistical Analyses*). Due to the nature of how trials will be triggered, we naturally expect variability across the number of trials that we will obtain in each run both within and across all subjects. Accounting for the variability of time taken to answer each thought probe, and the above timing parameters of trial onset across all trial types (brain-triggered or “time-out”), we anticipate collecting anywhere between 3 to 7 trials per run. While we do not expect as little as 3 trials to be captured during any given run, it would technically be possible to obtain only 3 trials in a run if all trials were “time-out” trials and the participant took the maximum amount of time possible to respond to each thought probe in each trial. We will use Psychophysics Toolbox in MATLAB R2020b to present our visual stimuli (Brainard, 1997) (see section *Real-time fMRI-triggered experience sampling*).

Statistical Analyses

The hypotheses that we will test in this study involve analyzing self-report experience-sampling ratings as a function of trial type (i.e., how an experience-sampling probe was triggered). Because trial type will depend on real-time fMRI analysis, the number of observations for each trial type will be unbalanced across participants. We consequently choose to test both hypotheses using linear mixed-effects models, which are particularly well-suited for estimating effects in designs where observations from participants are unbalanced across an experimental design (Pinheiro & Bates, 2000). All mixed effects models will be fit using the *lme4* package (Bates et al., 2015) in the *R* statistical software (R Core Team, 2022).

To test our first set of hypotheses (*H1*), **that subjective ratings of arousal time-locked to daIC-activation trials will be higher than ratings time-locked to non-daIC-activation trials (including both PMC-activation trials and time-out trials)**, we will model subjective ratings of arousal (response to Q3 prompt, “I felt very activated / energized.”) as a function of trial type. Trial type will be included in the model as a dummy-coded categorical fixed effect with two levels: (1) daIC-activation triggered probe (coded as 1); and, (2) non-daIC-activation triggered probe (coded as 0). Note that level (2) encompasses both PMC-activation triggered probes and time-out triggered probes, which were grouped together because both categories are defined by a lack of daIC activation. This dummy coded variable yields one slope parameter capturing the change in subjective ratings of arousal between daIC-activation triggered probes and non-daIC-activation triggered probes. We will include trial-wise head motion, estimated based on mean relative displacement estimated with FLIRT for the 5 TRs prior to thought probe onset, as a covariate to control for (see section *Criteria for Excluding Data*). To account for between-participant variability in the fixed effects, we will include the slope contrasting activation conditions, the model intercept, and all covariances between the fixed effects, as random effects grouped by participant. In doing so, we specify the maximal random effects structure allowed by the data, and we will iteratively simplify the model by dropping random effect parameters until (1) the model reaches convergence, and (2) the dropped random effect parameters prove to be significant based on likelihood ratio tests (LRTs; following prior recommendations (Barr et al., 2013)). We predict that the slope contrasting daIC-activation against PMC-activation and time-out trials will be significantly positive, suggesting that daIC-activation triggered trials are associated with higher levels of subjective arousal ratings relative to the other two trial types. We will estimate *p*-values for significance tests from well-established simulation methods using *lmerTest* in *R* (i.e., the

Satterthwaite degrees of freedom approximation; (Kuznetsova et al., 2017)), since assuming the null distribution of fixed effects parameters follows standard t or F distributions is often violated in practice (especially for unbalanced designs), and LRTs are known to produce anti-conservative p -value estimates (Pinheiro & Bates, 2000).

To test our second hypothesis ($H2$), **that subjective ratings of attention to inner mental experience time-locked to PMC-activation trials will be higher than ratings time-locked to non-PMC-activation trials (including both daIC-activation trials and time-out trials)**, we will conduct a similar linear mixed effects model as that reported above. However, instead of self-reported arousal, self-report attention to inner mental experience (response to Q1 prompt, “I was paying more attention to the task than my thoughts”) will be modeled as a function of trial type. Trial type will again be a dummy coded fixed effects categorical factor with two levels: PMC-activation triggered probe (coded as 1), and non- PMC- activation triggered probe (coded as 0). Similar to the previous hypothesis, non- PMC- activation triggered probe encompasses both daIC-activation triggered probes as well as time-out triggered probes. We will specify the model using the same methods described above: by specifying the maximal random effects structure allowed by the data and iteratively simplifying the model until convergence is reached and until random effects factors prove to be significant based on LRTs (Barr et al., 2013). We predict that the slope contrasting PMC-activation against daIC-activation and time-out trials ($H2$) will be significantly positive, suggesting that PMC-activation triggered probe trials are associated with higher levels of self-reported attention to inner mental experience relative to the other two trial types. All p -values will be estimated using the same methods described above for testing our first hypothesis.

Power Analysis

The proposed sample size of 60 was determined by conducting a power analysis on data from a previous study from our group that used random-interval experience sampling to assess mind-wandering (defined as task-unrelated thought, similar to Q1 in the current study) (Kucyi et al., 2016). These data consisted of 36 thought probes per subject, with DMN activity estimated from preprobe periods, for each of 28 participants. We assumed that the effect sizes we will observe in the present study (i.e., between activity in the ROIs and reports on fMRI-triggered experience sampling) will be at least as strong as that observed in this previous study. As noted, however, our rt-fMRI-ES paradigm is designed to enhance statistical power relative to random-interval experience sampling. Moreover, we will use functional localizers to target individual-specific ROIs during rt-fMRI-ES, a procedure that intends to enhance statistical power relative to prior work. As such, our power calculation may be viewed as conservative.

Data from the present study will be analyzed with a linear mixed model (see “Plans for Analysis” section below), and so, for consistency, we fit a linear mixed model to data from the previous study to estimate how many participants would be needed to achieve sufficient power. We modeled mind-wandering reports (0 to 100 scale with 100 being the most mind wandering) as a function of DMN percent signal change and included a random intercept and slope grouped by participant using the *lme4* package (Bates et al., 2015) in R (R Core Team, 2022). The unstandardized effect size was 23.425, suggesting that each unit increase in DMN percent signal change was associated with a 23.425 unit increase in self-reported mind wandering ($b = 23.425$, 95% CIs = [9.046, 39.964], $t(19,561) = 3.058$, $p = .006$). We used the fitted model to estimate power across a range of possible sample sizes using Monte Carlo simulations (1000 simulations each for sample sizes ranging from 3 to 45 in steps of roughly 5) using the *powerCurve* function

from the *simr* package (Green & MacLeod, 2016) with an alpha level set to 0.02. These simulations revealed that a sample size of 40 resulted in power levels of 0.857, sufficiently beyond the conventional 0.80 power threshold ($M = .857$, 95% $CI = [.834, .878]$; see **Fig 3**). To be conservative, we aim to recruit 20 extra participants to account for participants who fail to meet our criteria for inclusion in the study. This yields a total recruitment number of 60 participants.

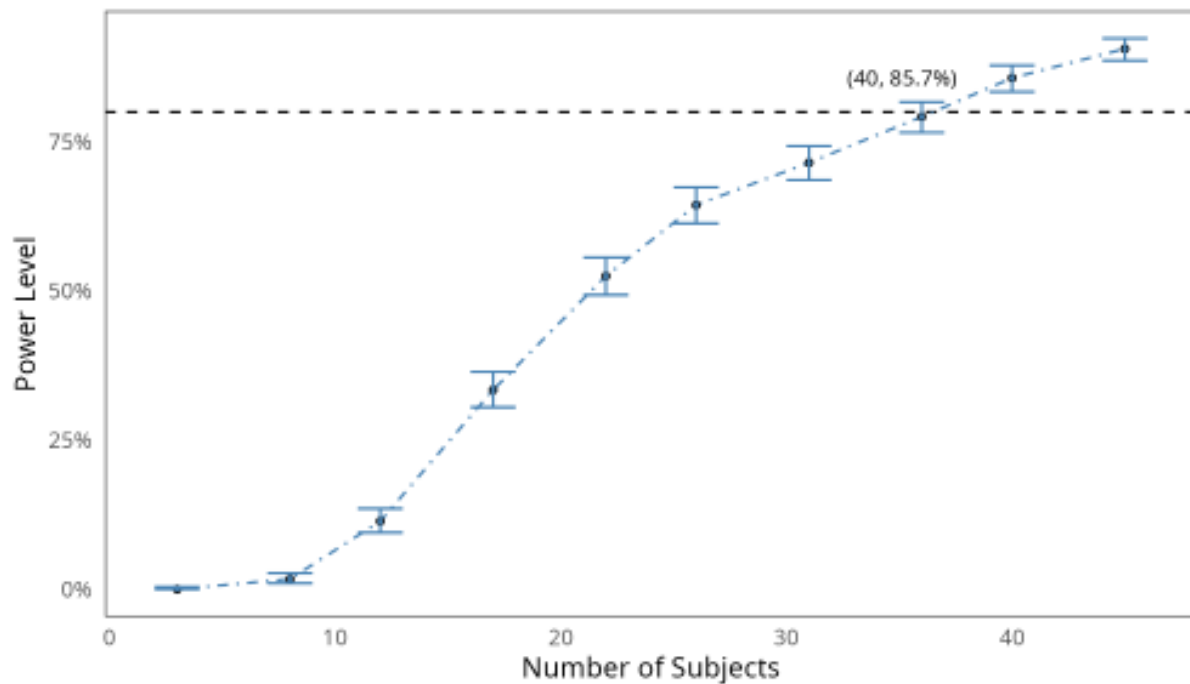


Figure 4. Power analysis. Estimated power level across sample sizes. Power was estimated from an effect size observed in previously collected data where subjective mind wandering reports were modeled as a function of DMN activity. Each point reflects the mean of 1000 Monte Carlo simulations estimating power level for a given sample size at an alpha level of 0.02. Error bars reflect 95% confidence intervals around means. Horizontal dashed line reflects conventional 80% power level threshold.

4. Criteria for Excluding Data

Following a review of the final collected dataset, we will exclude data due to excessive head motion, which could invalidate rt-fMRI analysis estimates and lead to erroneously triggered

experience-sampling prompts. Head motion will be assessed through both visual inspection of the fMRI images using the FSL tool *FSLeves* (McCarthy, 2021), and through motion estimates using FSL's *MCFLIRT* (Jenkinson et al., 2002). Following prior recommendations (Power et al., 2014), we will use frame-wise displacement (FD) as the main metric to be used for excluding trials and/or subjects. Furthermore, we will exclude any trials that are presented soon after (<10 seconds) the completion of the previous thought probe set. While our algorithm for triggering trials based on high-activation brain states was carefully written to mitigate quick consecutive trial onset (see section *Real-time fMRI Analysis*), a trial may occasionally follow another quickly after completion if the participant takes a long time to answer each thought probe throughout the trial. If a trial quickly follows another, the participant likely will not have had an ample amount of time to re-engage in attentional fluctuations toward and away from visual fixation. Furthermore, if a subject presses the pinkie button for the first thought probe of a trial indicating they were neither focused on the fixation cross nor their internal mental thoughts (i.e. distracted by external stimuli, or their mind was blank), we will exclude this trial. To account for lack of alertness during a trial, we will exclude any trials for which a subject recorded ratings for less than half of the eight thought probes. In summary, we will exclude individual trials when the following criteria are met:

- The presence of any volume with an FD value greater than 0.25 mm across the five volumes prior to onset of thought probes. Even though we will use data from three volumes to screen for events (see section *Real-time fMRI Analysis*), it is known that large head movements can impact subsequent frames (Power et al., 2014), and so we will perform exclusion based on five pre-probe volumes as a conservative measure.
- The trial appeared less than 10 seconds after the completion of a previous trial.

- For item 1 of the trial (Q1; see section *Real-time fMRI-triggered experience sampling (rt-fMRI-ES)*), the subject indicated they were neither focused on the fixation cross nor their internal mental thoughts.

- The subject recorded ratings for less than half of the thought probes in that trial.

Additionally, we will exclude all data from a subject when the following criteria are met:

- More than half of the subject's trials were excluded due to excessive head motion (based on trial-wise criteria specified above).
- More than 75% of the subject's trials were "time-out" events not triggered by BOLD activation (see section *Real-time fMRI-triggered experience sampling*).
- During more than 50% of the subject's trials, they were neither focused on the fixation cross nor their internal mental thoughts (see section *Real-time fMRI-triggered experience sampling (rt-fMRI-ES)*).
- During data acquisition, the subject was unable to complete more than 4 runs of rt-fMRI-ES.

To retain power for both of our proposed statistical tests (see section *Statistical Analyses*), we have imposed a minimum for both brain-triggered trial types (daIC-activation trials and PMC-activation trials) at 8 total observations for all participants. Thus, the following exclusion criteria will apply to all participant data for either statistical test:

- A participant will be excluded from the model testing the daIC hypothesis (H1) if the participant completed fewer than 8 daIC-activation triggered trials.
- A participant will be excluded from the model testing the PMC hypothesis (H2) if the participant completed fewer than 8 PMC-activation triggered trials.

We will not recruit new participants in place of those excluded post-data collection. Based on our power analysis, we concluded that a minimum sample size of 40 will be sufficient in retaining power for our statistical analyses (see section *Power Analysis*).

5. Acknowledgements

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6. PCI-RR Study Design Chart

Question	Hypothesis	Sampling plan	Analysis Plan	Rationale for deciding the sensitivity of the test for confirming or disconfirming the hypothesis	Interpretation given different outcomes	Theory that could be shown wrong by the outcomes
Does spontaneous BOLD activation in the daIC predict the occurrence of mental experiences that are characterized by high subjective arousal ?	H1: We predict that subjective ratings of arousal time-locked to daIC-activation trials will be higher than ratings time-locked to non-daIC-activation trials (including both PMC-activation trials and time-out trials). This hypothesis is rooted in evidence for a role of the daIC in assigning salience to spontaneous thoughts and in physiological arousal.	Power analysis was conducted using a previous study we conducted that used fMRI and random-interval experience sampling. Based on power simulations using the effect size we identified in that study, an N of 53 is required for .90 power. Given rt-fMRI-ES is intended to boost statistical power relative to random-interval experience sampling protocols, we consider this to be a conservative target sample size, and we will aim to collect data from 60 participants to account for participants who fail to meet our criteria for inclusion.	A linear mixed-effects model will be conducted with trial-types (daIC activation-triggered, PMC activation-triggered, or “time-out” non-brain-triggered) set as fixed effects, with random effects grouped by participant, and self-reported experience sampling arousal ratings as the outcome variable.	Based on prior fMRI literature inferring associations between spontaneous brain activity and experience sampling reports, setting the significance level at $P < 0.05$ will be sufficient to detect significant contrasts between the levels of our main predictor.	If daIC triggered trials are associated with significantly higher self-reported arousal ratings relative to the other two trial types, then our results will validate our rt-fMRI-ES technique by demonstrating its ability to capture salient spontaneous experiences using real-time BOLD analysis. This finding would further corroborate prior evidence that the daIC is linked to subjective arousal. If daIC activation triggered trials are not associated with significantly higher self-reported arousal ratings relative to the other two trial types, we	If our results support our hypothesis regarding daIC activation levels predicting high-arousal mind-wandering experiences, this finding would bolster insight into the role of non-default mode network brain regions in self-generated experiences. If our results do not support our hypothesis regarding daIC activation levels predicting high-arousal mind-wandering experiences, this finding would encourage further rigorous study into the salience network’s role in self-generated experiences. Such a finding could also motivate either the development of a more efficient approach of real-time BOLD

					will interpret this as a negative finding. These negative findings would constrain hypotheses to guide future work into refining real-time BOLD analysis, and studying the role of the daIC in subjective arousal.	analysis of the daIC, or the study of a different salience network node in association with mind-wandering.
Does spontaneous BOLD activation in the PMC predict the occurrence of internally-focused mind-wandering episodes?	H2: We predict that subjective ratings of attention to inner mental experience time-locked to PMC-activation trials will be higher than ratings time-locked to non-PMC-activation trials (including both daIC-activation trials and time-out trials). This is supported by a theorized role of the PMC in stimulus-independent thought, regardless of specific content.	Same as H1.	A linear mixed-effects model will be conducted with trial-types (daIC activation-triggered, PMC activation-triggered, or “time-out” non-brain-triggered) set as fixed effects, with random effects grouped by participant, and self-reported experience sampling attention ratings as the outcome variable.	Same as H1.	If PMC activation triggered trials are associated with significantly higher self-reported internal attention ratings, then our results will validate our rt-fMRI-ES technique by demonstrating its ability to capture the onset of self-referential spontaneous experiences using real-time BOLD analysis. This finding would further corroborate prior evidence that the PMC is linked to internally-focused cognition. If PMC activation	If our results support our hypothesis regarding PMC activation levels predicting increased internally-focused attention, this finding would build upon previous work connecting DMN regions to mind-wandering via retrospective data analyses, offering plausible real-time fMRI evidence that mind-wandering episodes can be prospectively predicted from ongoing brain activity. If our results do not support our hypothesis regarding PMC activation levels

					triggered trials are not associated with significantly higher self-reported internal attention ratings, we will interpret this as a negative finding. These negative findings would constrain hypotheses to guide future work into refining real-time BOLD analysis, and studying the role of the PMC and other DMN nodes in internally-focused attention.	predicting increased internally-focused mind-wandering experiences, this finding would encourage further rigorous study into the PMC's role in self-generated experiences. Such a finding could motivate either the development of a more efficient approach of real-time BOLD analysis of the PMC, or the study of a different DMN node in association with mind-wandering.
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